25901158 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECCHE200 - BASIC ANALYTICAL AND COSMETIC CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very short answer questions

Answer any **10** questions

Each question carries **2** marks

1. Explain the advantages of hand sanitizer. [U] / [CO1]
2. What is the function of preservatives in hand wash formulation ? [U] / [CO1]
3. Differentiate between acetone and non-acetone nail polish removers. [U] / [CO1]
4. Describe the factors affecting nail growth. [K] / [CO1]
5. How are soaps classified based on their TFM values? [K] / [CO2]
6. What is the active ingredient in clove used in oral care? [A] / [CO2]
7. How soaps are classified based on their PH range. [U] / [CO2]
8. Name any oil used in soap manufacture. [U] / [CO2]
9. What is the role of **soil minerals** in soil composition? [U] / [CO3]

10. What is meant by RO (Reverse Osmosis) purification? [U] / [CO3]

11. Write the chemical formula of water and name its constituent elements. [U] / [CO3]

12. Write any two applications of soil pH analysis. [U] / [CO3]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

13. Discuss how hand care products help maintain skin health. [U] / [CO1]

14. Explain the basic steps involved in making a hand wash. [U] / [CO1]

15. Define nail polish. Describe the meanings and significance of the terms nail lacquer and nail enamel. [U] / [CO1]

16. Discuss the role of humectants, binders, and surfactants used in toothpaste. [A] / [CO2]

17. Discuss the procedure for the preparation of cream soaps. [K] / [CO2]

18. Explain the role of essential oils in improving therapeutic and sensory properties of oral care products. [A] / [CO2]

19. Describe the different water sampling methods used for testing water quality? [U] / [CO3]

20. Describe the practical procedure for analysis of soil pH and explain how the results are used in agriculture. [K] / [CO3]

[5x6 = 30]